Applied Energy 285 (2021) 116485

Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apenergy

A hybrid model for carbon price forecasting using GARCH and long short-term memory network

Yumeng Huang [a], Xingyu Dai [b], Qunwei Wang [b,*], Dequn Zhou [b]

[a] College of Environmental Science and Engineering, Peking University, Beijing 100871, China
[b] College of Economics and Management, Nanjing University of Aeronautics and Astronautics, Nanjing 211106, China

HIGHLIGHTS

- Propose a novel carbon price forecasting model adapted to the EU ETS post-2017.
- Combine econometric and neural network methods comprehensively.
- Decompose series through variational mode decomposition for multiscale prediction.
- Use nonlinear ensemble to narrow the gap between mode forecasts and original series.
- Show superiority in forecasting accuracy and robustness over benchmark models.

ARTICLE INFO

Keywords:
Carbon price forecasting
Variational mode decomposition
GARCH
LSTM neural network

ABSTRACT

The reform of the EU ETS markets in 2017 has induced new carbon price forecasting challenges. This study proposes a novel decomposition-ensemble paradigm VMD-GARCH/LSTM-LSTM model to better adapt to the current fast-rising and volatile carbon price. Three significant steps are involved: (1) the Variational Mode Decomposition (VMD) algorithm decomposes the carbon price series into sub-modes; (2) The Long Short-Term Memory (LSTM) network predicts low-frequency sub-modes, with the GARCH model predicting high-frequency sub-modes; (3) the forecasts from sub-modes are ensembled through the LSTM non-linear ensemble method. Combining econometric and artificial intelligence methods, our proposed model has an excellent performance on the current carbon price, with smaller errors than single econometrics or AI models or decomposition-ensemble models with linear simple superposition approaches. VMD have significant advantages over their alternative algorithms. Moreover, the LSTM involved in our model is well suited to forecast the rising carbon price in late EU ETS Phase III, providing good insight into risk aversion for participants.

1. Introduction

European Union Emissions Trading System (EU ETS), the largest ETS market globally [1], provides an environmental, financial instrument market to encourage power stations and industrial plants, among other massive energy-using installations to use clean energy by issuing and selling carbon emission permits. This leads directly to the widespread interest in carbon price forecasting, such as European Union Allowance (EUA), the most widely noticed varieties of carbon credits, by energy market participants and speculative investor for risk aversion and hedging (see among [2,3,4,5,6,1] with others). This study proposes a new forecasting approach to better address the current physical carbon market conditions.

The work of forecasting EUA prices in its late Phase III is challenging now. Compared to the previous two Phases (Phase I, 2005–2008 and Phase II, 2008–2013), the implementation of the Market Stability Reserve (MSR) and Linear Reduction Factor (LRF) policy has significantly reduced the supply of carbon credits, so that the carbon price is no longer stuck at a low level [7,8,9]. After 2017, increasing enforcement on LRF and MSR has further deepened the magnitude of the carbon price variation, and the EU ETS carbon market shows a brand new feature[1]. The carbon price has climbed up to €35 from €5 at the beginning of 2017

* Corresponding author.
E-mail address: wqw0305@126.com (Q. Wang).

[1] EUA prices were in a very depressed state until 2017 due to the large number of carbon permits issued. A glut relative to demand of roughly 1.8 billion allowances, which exists in the form of banked quantities on firm's accounts before 2015 [28].

https://doi.org/10.1016/j.apenergy.2021.116485
Received 17 November 2020; Received in revised form 5 January 2021; Accepted 11 January 2021
Available online 22 January 2021
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Table 1
Literature review on carbon price forecasting.

Authors	Products	Timespan	Decomposition	Econometric	Artificial intelligence
Paolella and Taschini [35]	EUA spots	Phase I	/	Mixed-normal GARCH	/
Benz and Trück [36]	EUA	Phase I	/	MS-GARCH	/
Daskalakis et al. [39]	EUA futures	Phase I & II	/	SPDE	/
Chevallier [32]	EUA spots	Phase II	/	Factor-Augmented VAR	/
Chevallier [62]	EUA spots	Phase I & II	/	Non-parametric model	/
Guðbrandsdóttir and Haraldsson [33]	EUA spots	Phase I & II	/	Linear regression	/
Chevallier and Sévi [38]	EUA futures	Phase II (2008)	/	HAR-RV	/
Zhu [13]	EUA futures	Phase I & II	EMD	/	ANN + Optimization Algorithm
Koop and Tole [11]	EUA spots	Phase I & II	/	Time-varying linear regression	/
Byun and Cho [3]	EUA futures	Phase I & II	/	GARCH	/
García-Martos et al. [10]	EUA and CER spots	Phase II	/	VARMA	/
Zhu and Wei [1]	EUA futures	Phase II	/	ARIMA	LSSVM + Optimization Algorithm
Li and Lu [25]	China carbon spot	From 2013 to 2014	EMD	GARCH	/
Fan et al. [18]	EUA futures	Phase II & III	/	/	FNN
Atsalakis [2]	EUA futures	Phase II	/	/	ANFIS
Sun et al. [12]	EUA futures	Phase II	VMD	/	SNN
Zhu et al. [28]	EUA futreus	Phase II & III	EEMD	ARIMA	LSSVM + Optimization Algorithm
Li et al. [40]	EUA spots	Phase I & II	/	Regime-switching jump diffusion model	/
Zhu et al. [6]	EUA futures	Phase III	EMD	/	LSSVR + Optimization Algorithm
Zhang et al. [5]	EUA spots	Phase I & II & III	CEEMD	Co-integration + GARCH	GNN + Optimization Algorithm
Zhao et al. [34]	EUA futures	Phase III	/	MIDAS + Forecast combination method	/
Sun and Zhang [31]	EUA futures	Phase II	MRSVD	/	NN + SVM + ELM + Optimization Algorithm
Zhu et al. [26]	EUA	Phase III	EMD	GARCH	LSSVM
Tian and Hao [4]	EUA futures	Phase I & II and some Phase III	VMD	/	ANN + Optimization Algorithm
Xu et al. [19]	EUA futures	Phase II & III	/	/	Complex network + ELM
Hao and Tian [29]	China carbon spot	From 2014 to 2018	ICEEMDAN	/	KELM + Optimization Algorithm
Lu et al. [30]	China carbon spot	From 2013 to 2019	CEEMDAN	/	NN + SVM + Optimization Algorithm

and has fluctuated considerably.

Although some classical and fascinating work has helped in the prediction of the carbon price during Phase I & Phase II, and even early Phase III (see among [2,10,11,12,13] with others), the prices during that period were too low to accommodate current market characteristics, and did not adequately help investors predict current carbon price characteristics. A carbon price forecasting study for the EU ETS in Phase III, especially after the MSR policy is activated, is much needed by investors right now.

Besides, the non-linear features that have been found for carbon price (see among [14,15,16,17] with others), which implies that the traditional forecasting methods, such as ARIMA, GARCH or other linear models, may not be a good choice[2]. Many studies have used AI methods to forecast carbon prices [2,18,19]. However, in the face of the current highly volatile carbon market, these "black box" AI methods' predictive effectiveness is a cause for concern[3].

Following the philosophy of "divide and conquer," the decomposition-ensemble forecasting method has attracted much attention in recent years[4] [20,21,22]. Due to the heterogeneity of financial product prices over different time scales [23,24], the decomposition-ensemble forecasting method has been applied to various fields [21,22] and carbon markets (see [25,12,13,26] among others). However, few decomposition-ensemble model-based studies take advantage of both econometric and artificial intelligence approaches and explore the current EU ETS Phase III carbon market[5]. All of these facts are strongly driving us to forecast the current carbon market in a more appropriate way to help investors better anticipate risk in today's highly volatile carbon price.

This study constructs a more comprehensive approach to predicting the current highly volatile carbon price to solve this problem. We propose a decomposition-ensemble carbon price forecasting model named VMD-GARCH/LSTM-LSTM, whereby the model plays to the strengths of both econometric and artificial intelligence models. We first decompose the carbon price series into different frequency sub-modes through variational modal decomposition (VMD) of Dragomiretskiy and Zosso [27], then develops Generalized autoregressive conditional heteroskedasticity (GARCH) and Long Short-Term Memory Model (LSTM) for high- and low-frequency sub-modes respectively, finally combines forecasts of all the modes by the LSTM non-linear ensemble algorithm.

The contributions made by this study are as follows. Firstly, we specifically designed the VMD-GARCH/LSTM-LSTM model based on the decomposition-ensemble methodology for the new carbon market

[2] Evidence from previous studies have also shown that the EUA price is regime-switching within a Phase period [36]. Moreover, Tan and Wang [43] and Feng et al. [19] reveal that the tails of the carbon price distribution can be affected by other energy products.

[3] Existing econometric parametric models do not adequately account for all the non-linear features of the carbon price series, and "black box" methods such as machine learning often do not perform well in the face of complex time series. For a detailed discussion, see Section 2.

[4] The method decomposes the price sequence into sub-modes with specific frequency bands or periodicity by a decomposition method, and then applies a specific prediction model to each modality. The predicted sub-modes are then merged to obtain the predicted value of the original series.

[5] For a detailed discussion, see Section 2.

scenario of a dramatic increase in the EU ETS after 2017. The applicability of previous carbon price forecasting studies to the current carbon market is likely to be uncertain due to the dramatic carbon market changes. When using carbon price data after 2017 for verification and testing, our model VMD-GARCH/LSTM-LSTM is superior to various benchmark models in two sample cases, indicating the proposed paradigm's effectiveness and the model combination's accuracy and robustness. Our research drives the evolution of carbon price forecasting in a new temporal and spatial context. We seek to provide a useful forecasting tool for current and future carbon market participants.

Secondly, compared to existing studies that use the decomposition-ensemble method for carbon price prediction, we have effectively improved the prediction method. In contrast to previous studies that mostly used EMD or EEMD for modal decomposition for carbon price prediction[25,13,28,6], we employ a novel VMD model for decomposition, thus overcoming the inherent limitations of end effect and mode mixing problems in EMD or EEMD. The study results provide ample evidence that the use of the VMD model leads to an improvement in prediction accuracy compared to the EMD model.

Thirdly, in contrast to decomposition-ensemble studies using AI or econometric methods alone [29,25,30,31,12,4,13,6], we used a combined GARCH and LSTM model for sub-modes of different frequencies. As the carbon market is currently on the upswing, long-term trends may contain important predictive information. The LSTM model for sub-modes with long periods can take full advantage of its long memory. The memory advantage of the LSTM model for long-term trends in the post-2017 carbon price is unmatched compared to other traditional AI models [2,18,13,1]. The use of the GARCH model for short-period sub-modes can better exploit the GARCH model's depiction of cluster volatility in our VMD-GARCH/LSTM-LSTM framework. The LSTM non-linear ensemble technique is introduced to improve the prediction performance further. Our empirical results fully reflect the advantages of ensemble using LSTM, a finding that drives improvements in applying traditional simple summation ensemble for carbon price forecasting (see [31,5] among others).

The remaining parts of this paper are organized as follows. Section 2 reviews the studies on carbon price forecasting and reveals the research gaps. Section 3 presents the formulation process of the proposed model. Section 4 describes the source data. Section 5 discusses the empirical results. Section 6 concludes the paper and outlines further research.

2. Literature review on carbon price forecasting

Table 1 vividly presents the literature review. Different attempts have been made in the previous literature to overcome the effect of non-linearity on the effectiveness of carbon price prediction. Several studies have used univariate or multi-variate linear models to predict the carbon price, including ARIMA, Factor-Augmented VAR, VARMA model time-varying parameter regression and MIDAS linear regression [32], [11], [33], [10] and [34]. As noted in the Introduction section, the volatility of the carbon price and the tail characteristics still drive the construction of the carbon price using GARCH or other volatility models to overcome features such as the spike-thick-tailed nature of the carbon price distribution.

In this vein, Paolella and Taschini [35], Benz and Trück [36], Lutz et al. [37], and Byun and Cho [3] confirmed the excellent predictive performance of different GARCH-types models during Phase I and II. Chevallier and Sévi [38] used high-frequency carbon price data during Phase II to make volatility predictions. The jump, a highly non-linear characterization, has also been noticed by the field of carbon price forecasting [39,40].

With big data and computer technology development, machine learning is widely used in carbon price prediction in EU ETS Phase II [2,13,18,19]. Fan et al. [18] use FNN to forecast EUA futures in EU ETS Phase II and the beginning of Phase III. ANFIS method is used in Atsalakis [2]. Many studies consider combining econometric and AI methods to take advantage of their carbon price forecasting strengths to overcome the drawbacks of AI methods. A classic study of Zhu and Wei [1] combines the SVM with an ARIMA model and uses an optimization algorithm to predict the price of EUA futures during Phase II.

With the rapid development of decomposition–ensemble forecast methods, carbon price forecasting techniques have shown a new face in recent years. Several studies have decomposed the original carbon price series using different methods and predicted the subsequences using different models to obtain the original series' predicted values, as described in the introduction section. Zhu [13] employs the EMD decomposition method and ANN method with an optimization algorithm method to forecast EUA futures in Phase III, indicating that the decomposition-ensemble method outperforms the simple ANN forecast ability on the carbon price. A similar finding is confirmed by Zhu et al. [6], where EMD and LSSVM are used to predict EUA futures price at Phase III. Li and Lu [25] confirm EMD and GARCH's good performance on China carbon price forecasting.

As decomposition methods continue to advance, several more data-adaptive decomposition methods have been adopted and combined with econometrics and machine learning methods. Applying variational modal decomposition (VMD) and SNN method, Sun et al. [12] shows the advantages of VMD decomposition for carbon price prediction. Tian and Hao [4] combines the VMD and ANN to forecast the EUA futures. Hao and Tian [29] predicted carbon prices in different carbon markets in China using a kernel extreme learning machine model. Using the KELM method, they found that the original series' prediction after ICEEMDAN decomposition was significantly better with EMD and EEMD. Lu et al. [30] applied the CEEMDAN decomposition method, NN, and SVM to forecast 8 China carbon markets. More decomposition–ensemble model applied in carbon price forecasting can be seen in Sun and Zhang [31], Zhang et al. [5], and Zhu et al. [26,28].

Going back to previous studies, we find the following latent research gaps in the carbon price forecasting, as shown in Table 1. (1) Most of the forecast studies stayed in the EU ETS Phase I versus Phase II period or in non-EU ETS markets. Despite these studies' success with Phase I and Phase II, however, the carbon market's current complexity during late Phase III may allow for non-linearity between other variables and the carbon price, thus modeling misspecification. This raises concerns about the ability of Phase III to predict the highly volatile carbon price well. (2) There are advantages and disadvantages to using either metrics or machine learning methods alone, and a predictor that combines both methods is worth being expected to be used for the EU ETS late Phase III carbon price study. The GARCH-type models portray the volatility information to give more accuracy on carbon price forecasting. However, it is difficult for these traditional econometric and statistical methods to capture the nonlinear characteristics hidden in carbon price. (3) The application of decomposition-ensemble prediction methods is not yet comprehensive, and most of them only apply metrics or machine learning methods to decomposed carbon returns. Few studies have applied econometric and machine learning methods comprehensively to carbon price prediction for subseries data features. (4) Few decomposition-ensemble carbon price forecasting studies have used non-linear patterns to integrate predicted submodel sequences. Nevertheless, the use of a simple additive total ensemble approach lacks a supporting basis. To fill gaps, our study uses VMD decomposition and the combined application of GARCH and machine learning methods, that is, LSTM, to predict the carbon price during EU ETS late Phase III.

3. Methodology

3.1. VMD algorithm

The VMD algorithm was proposed by Dragomiretskiy and Zosso [27] and is an improvement based on EMD. Traditional decomposition techniques have strict limitations on the data, for instance, Fourier transformation is suitable for smooth periodic signals [41], and wavelet

analysis can effectively capture the transient effect in the signal under the assumption of non-stationary and linear data [42]. By contrast, with good flexibility and adaptability, EMD can capture complex signal features without a priori information. Despite the merits, EMD also has limitations, such as end effect and mode mixing, which restrain the decomposition performance. Thus, VMD was proposed to remedy these limitations.

The target of VMD is to decompose the input signal y into integer number K sub-signals (modes), and each mode is a limited bandwidth with central frequency w_k. The calculation steps of the mode bandwidth are as follows:

(1) For each mode y_k, perform the Hilbert transformation, and construct an analytical signal to obtain a unilateral spectrum.
(2) By mixing with the exponent tuned to the corresponding estimated central frequency, the frequency spectrum of y_k is tuned to the baseband.
(3) The bandwidth of $y_k$is estimated from the H1 Gaussian smoothness of the demodulated signal.

VMD can be written as the following constraint variation problem:

$$\min_{\{y_k\},\{\omega_k\}}\left\{\sum_{k=1}^{K}\left\|\partial_t\left[\left(\delta(t)+\frac{j}{\pi t}\right)*y_k(t)\right]e^{-j\omega_k t}\right\|_2^2\right\} \tag{1}$$

subject to

$$\sum_{k=1}^{K} y_k = y \tag{2}$$

where y is the signal to be decomposed, and $\{y_k\}=\{y_1,y_2,\cdots,y_K\}$ and $\{\omega_k\}=\{\omega_1,\omega_2,\cdots,\omega_K\}$ represent the set of all modes and the corresponding central frequency, respectively. δ is the Dirac distribution, K is the number of modes, and * is convolution, and j is $\sqrt{-1}$.

By introducing a penalty term and Lagrange multiplier, the aforementioned constraint problem can be transformed into an unconstrained problem that is easier to solve:

$$L(\{y_k\},\{\omega_k\},\lambda)=\alpha\sum_{k=1}^{K}\left\|\partial_t\left[\left(\delta(t)+\frac{j}{\pi t}\right)*y_k(t)\right]e^{-j\omega_k t}\right\|_2^2$$
$$+\left\|y(t)-\sum_{k=1}^{K}y_k(t)\right\|_2^2+\left\langle\lambda(t),y(t)-\sum_{k=1}^{K}y_k(t)\right\rangle \tag{3}$$

where α is a balance parameter of the data fidelity constraint.

Eq. (3) is solved by the suboptimal iterative sequence named alternating direction multiplier method, namely, updating y_k and ω_k in two directions. The results are as follows:

$$\hat{y}_k^{n+1}(\omega)=\frac{\hat{y}(\omega)-\sum_{i\neq k}\hat{y}_i(\omega)+\frac{\hat{\lambda}(\omega)}{2}}{1+2\alpha(\omega-\omega_k)^2} \tag{4}$$

$$\omega_k^{n+1}=\frac{\int_0^\infty \omega|\hat{y}_k(\omega)|^2 d\omega}{\int_0^\infty |\hat{y}_k(\omega)|^2 d\omega} \tag{5}$$

where $\hat{y}(\omega)$, $\hat{y}_i(\omega)$, $\hat{\lambda}(\omega)$, and $\hat{y}_k^{n+1}(\omega)$ represent the Fourier transformation of $y(t)$, $y_i(t)$, $\lambda(t)$, and $y_k^{n+1}(t)$, respectively, and n is the number of iterations.

3.2. LSTM neural network

As the price of the EU ETS carbon market is in a period of rapid price increase after 2017, the price's active pattern is significantly different from the past, and a model that is well able to assimilate information about this long-term upward trend entirely is worth considering. Long short-term memory (LSTM) proposed by Hochreiter and Schmidhuber [43] is a special kind of RNN that is designed to solve the problem of gradient disappearance and gradient explosion during long sequence training. In short, LSTM can adequately retain the long-term variation pattern of the sequence within the parameters [44], which is very suitable for depicting the long-term rising carbon price pattern after 2017.

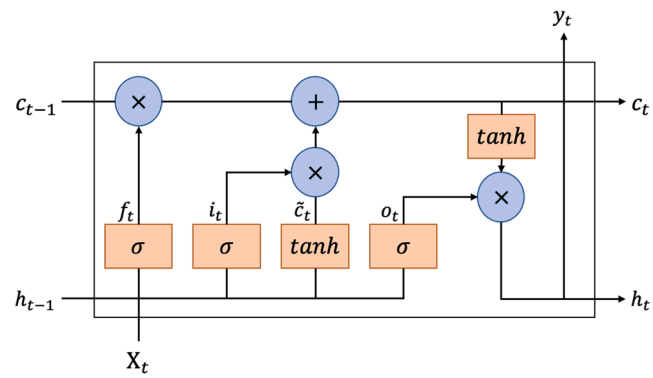


Fig. 1. Internal structure of the LSTM unit.

The long-period mode after VMD decomposition has a obviously trend component[45]. LSTM can better absorb this long-period trend component than by other artificial intelligence methods such as support vector machines (SVM). Moreover, the long-run trend component of the carbon price series is highly non-linear and non-stationary, which is not suitable for fitting with general econometric models.

An LSTM unit contains two hidden states, namely, h_t and c_t, for storing short-term information and long-term information, respectively. Three control gates—forget gate, input gate, and output gate—are also introduced. The internal structure of the LSTM unit is shown in Fig. 1.

The forget gate f_t determines how much information is saved from the previous state c_{t-1}. The forget gate at time t is

$$f_t=\sigma\left(\boldsymbol{W}_f\cdot[h_{t-1},X_t]+b_f\right) \tag{6}$$

where $\sigma(\cdot)$ represents the sigmoid activation function, and X_t represents the input of the prediction model, namely, historical carbon price data. f_t represents the forget gate vector at time t; h_{t-1} represents the output vector at time $t-1$ (also the state h vector); $\boldsymbol{W}_f$ and b_f, respectively, represent the weight matrix and bias of the forget gate; $[\cdot]$ indicates a vector concatenation operator.

The input gate i_t determines how much current information should be input to produce the current state c_t. i_t is calculated as follows:

$$i_t=\sigma(\boldsymbol{W}_i\cdot[h_{t-1},X_t]+b_i) \tag{7}$$

where $\boldsymbol{W}_i$ and b_i represent the weight matrix and bias of the input gate, respectively. We observe that i_t and f_t have a similar form, and both gates are determined by h_{t-1} and X_t.

The current hidden state c_t is obtained by adding the information controlled by the two gates f_t and i_t. Long-term information is controlled by f_t, and short-term information is controlled by i_t:

$$c_t=\tanh(\boldsymbol{W}_c\cdot[h_{t-1},X_t]+b_c) \tag{8}$$

$$c_t=f_t*c_{t-1}+i_t*c_t \tag{9}$$

where $\tanh(\cdot)$ represents the tanh activation function, $\boldsymbol{W}_c$ and b_c respectively represent the weight matrix and bias of the current state, and operator * represents the element-wise product, that is $\begin{bmatrix}a_{11} & a_{12}\\ a_{21} & a_{22}\end{bmatrix}*\begin{bmatrix}b_{11} & b_{12}\\ b_{21} & b_{22}\end{bmatrix}=\begin{bmatrix}a_{11}b_{11} & a_{12}b_{12}\\ a_{21}b_{21} & a_{22}b_{22}\end{bmatrix}$

The last task of the LSTM unit is to calculate how much information should be output, which is completed by the output gate o_t:

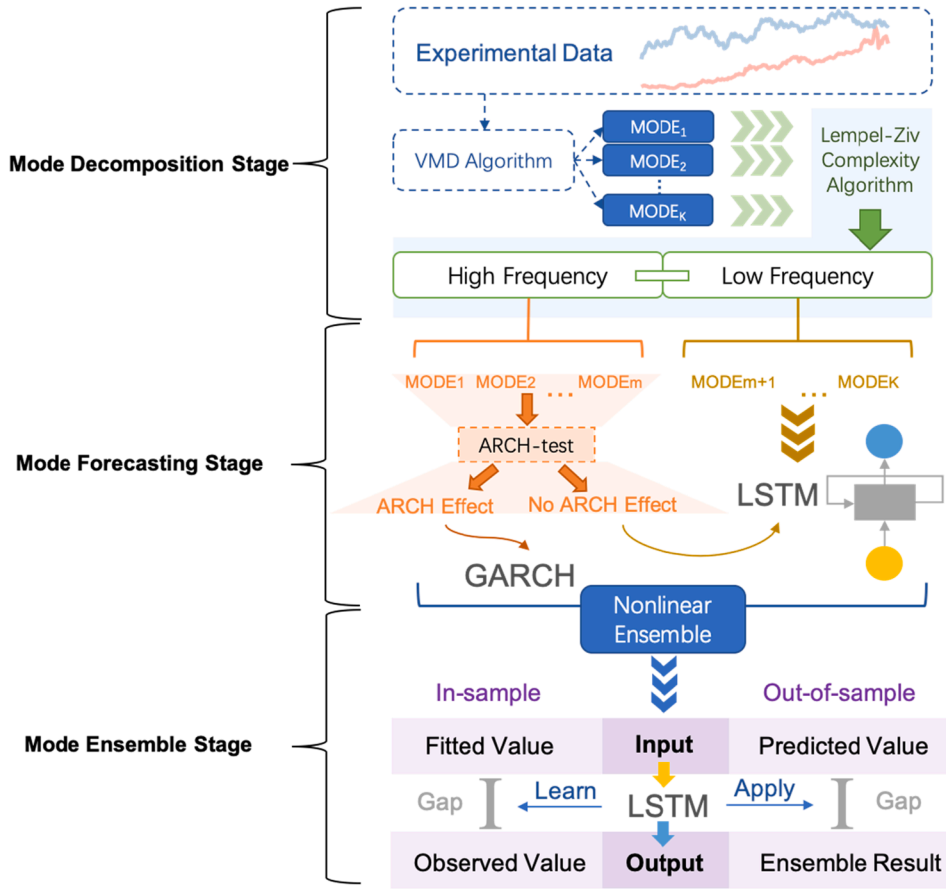


Fig. 2. Carbon price forecasting process of VMD-GARCH/LSTM-LSTM.

$$o_t = \sigma(\boldsymbol{W}_o \cdot [h_{t-1}, X_t] + b_o) \tag{10}$$

The final output of the LSTM unit is

$$h_t = o_t * \tanh(c_t) \tag{11}$$

3.3. GARCH model

The generalized autoregressive conditional heteroskedasticity (GARCH) model of Bollerslev [46] is a classical model for depicting cluster volatility. The current conditional variance depends on l previous errors and k previous conditional variances:

$$v_t = \varsigma_0 + \sum_{i=1}^{k} \varsigma_i v_{t-i} + \sum_{i=1}^{l} \eta_i \varepsilon_{t-i}^2 \tag{12}$$

where ς_i is a constant coefficient, and the error term ε_t follows a GARCH process of order k and l, written as GARCH(k,l). ς_0, ς_i, and η_i are nonnegative. GARCH models have been used to fit carbon price fluctuations [36,3,35]. An important realization is that the GARCH model is only applicable to stationary series and that the data are highly fluctuating around a mean value. The high-frequency sub-modes decomposed by the VMD method are very volatile without any trend component (i.e., the sequence is highly smooth of I(0)) due to their very short period. Sub-modes with this characteristic are well suited to be fitted with GARCH [26].

The fluctuation patterns of the high-frequency sub-modes decomposed using the VMD model are more stable (this evidence can be continued in Section 5.1), implying that our use of the GARCH model is sufficient to adequately depict their variations [26], rather than having to use some asymmetric GARCH model such as EGARCH, GJR-GARCH or TGARCH.

3.4. VMD-GARCH/LSTM-LSTM model

The multi-scale non-linear ensemble learning model for carbon price forecasting proposed in this paper, namely, VMD-GARCH/LSTM-LSTM, deftly integrates the VMD, GARCH, and LSTM models. We divided the model construction process into three stages (Fig. 2).

3.4.1. Stage 1: Mode decomposition and frequency component identification

The carbon price is decomposed into K modes with different amplitudes and frequencies, arranged from high frequency to low frequency. The modes are independent of each other, with simple structures, stable fluctuations, and strong regularity.

Modes are classified into two categories according to frequency: (1) The high-frequency component has a high frequency and a low amplitude and characterizes the random changes in the carbon price caused by the imbalance between supply and demand normal operation of the short-term market. Although the high-frequency component frequently fluctuates in the short term, it cannot produce long-term effects. Moreover, wild fluctuations in the carbon price are often followed by wild ones, and minor fluctuations are followed by minor ones, namely, volatility clustering exists. (2) The low-frequency component has a low frequency and a high amplitude and captures carbon price changes caused by the external environment. The impact of a major accident causes the value of the low-frequency component to rise or fall rapidly. Despite the low frequency of fluctuations, the component may cause dramatic changes in the carbon price or even change its price mechanism.

According to the analysis above, different frequency components have different data characteristics. If the forecasting model for each component is selected pertinently, the accuracy of carbon price

prediction can be improved. To identify frequency components, we use the Lempel–Ziv complexity algorithm [47] after VMD decomposition to specify the optimal model to predict each component [26]. The Lempel-Ziv algorithm measures the rate at which new patterns emerge in a time series, and the higher the LZ value it calculates, the faster the new patterns emerge. Since high-frequency sequences are mostly short-period oscillations, their changing patterns will repeat in a short time, while the opposite is true for long-period sequences. Thus, the LZ value of high-frequency sequences (short period) should be much smaller than that of low-frequency sequences (long period). We thus use the LZ value to divide the high-frequency and low-frequency sequences. The identification process is as follows:

Step 1: calculate the complexity C_k of mode k, $k = 1, 2, \cdots, K$.

Step 2: set the critical value $\lambda_0 (\geq 0.8)$ and find the minimum value of m that satisfies the following formula:

$$\eta = \frac{\sum_{k=1}^{m} C_k}{\sum_{k=1}^{K} C_k} \geq \lambda_0, m \in k \tag{13}$$

Step 3: modes 1 to m are identified as the high-frequency component; modes m + 1 to K are identified as the low-frequency component.

3.4.2. Stage 2: Prediction of high-frequency and low-frequency components

We choose the appropriate modeling methods according to different data characteristics of high-frequency and low-frequency components and develop a forecasting model for each mode separately; the details of which are as follows:

(1) Prediction of the high-frequency component. The high-frequency component exhibits short-term fluctuations in carbon price and is characterized by high randomness, time-variation, and clustering. First, if the ARCH effect exists, a GARCH forecasting model is constructed; otherwise, an LSTM forecasting model is constructed. As mentioned above, GARCH is an econometric model that has strong short-term memory capabilities for stochastic processes with time-varying and clustering features and is suitable for predicting the high-frequency component [25,26].

(2) Prediction of the low-frequency component. LSTM is suitable for processing and predicting important events with very long intervals and delays in the time series [43]. After 2017, the EU ETS carbon price shows a strong upward trend. The period of the low-frequency carbon price sequence after VMD decomposition is very long. Because the low-frequency component has strong periodic fluctuation characteristics and clear change patterns, the LSTM model with excellent non-linear modeling capabilities is used for prediction. As described in Section 2.2, LSTM can store long-term carbon price series information by constructing memory units.

3.4.3. Stage 3: non-linear ensemble of frequency component forecasts

The predicted values of high-frequency and low-frequency components are ensembled to achieve the final carbon price forecasting result. We adopt the non-linear ensemble technique, namely, training an LSTM ensemble model, which takes the sum of mode predictions at each time point in the training set as the input, and the real value of carbon price at that point as the output, to learn the relationship between predicted values and real values. For each sum of the mode predictions in the test set, apply the ensemble model to modify the forecasting error and obtain the final result.

Notably, to avoid the data leakage problem which is detailed in Section 4.1, after an out-of-sample prediction is made, the aforementioned three stages must be repeated to predict the next data point.

3.5. Prediction evaluation criteria and statistical tests

To compare the forecasting performance of different models, we must choose loss fuctions and statistical tests. We employ three popular loss functions of root mean square error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE), which three has been widely used in price forecasting studies [31,26,1]. Using multiple loss functions indicators, it may not be fair to evaluate by a single indicator, as some models rank first under the A-indicator evaluation system and second under the B-indicator ranking system.

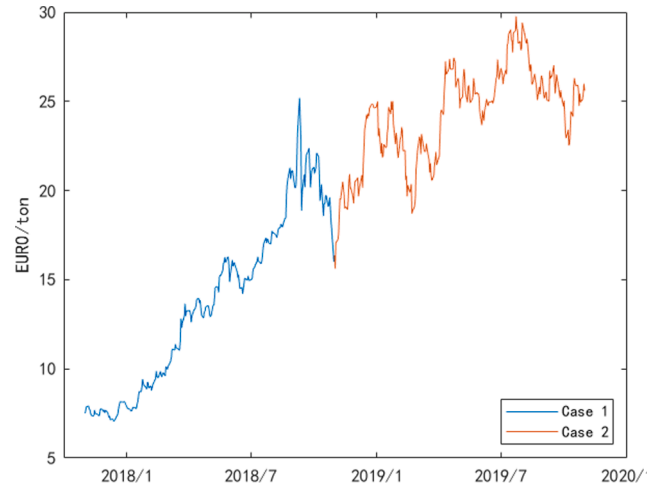


Fig. 3. Carbon price sample cases division.

RMSE, MAE, and MAPE are defined as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n} [x(t) - \hat{x}(t)]^2} \tag{14}$$

$$MAE = \frac{1}{n}\sum_{t=1}^{n} |x(t) - \hat{x}(t)| \tag{15}$$

$$MAPE = \frac{1}{n}\sum_{t=1}^{n} \frac{|x(t) - \hat{x}(t)|}{x(t)} \tag{16}$$

where $x(t)$ and $\hat{x}(t)$ are the real value and predicted value of carbon price at time t, respectively, and n is the test set's sample size.

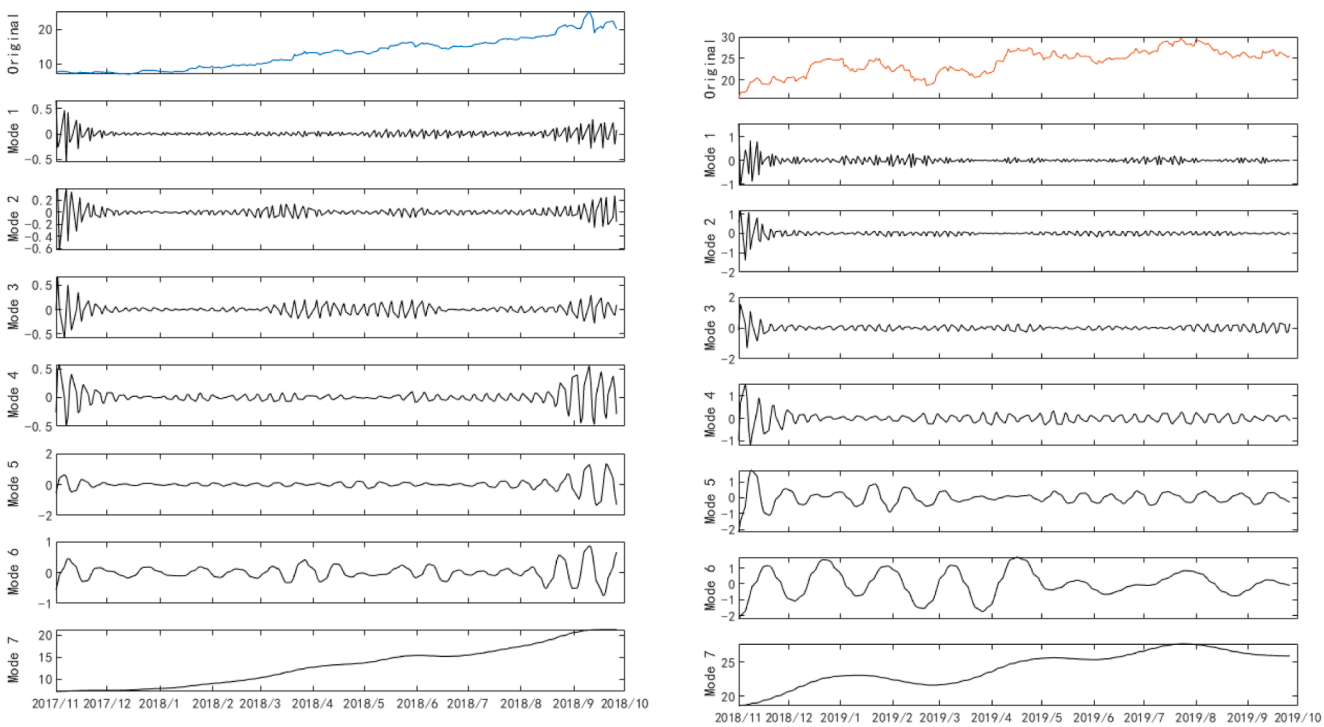
One concern is that when most of the predictions for a particular model are very accurate, a particular outlier prediction may make the loss function perform very poorly. In addition, the loss function can be viewed as a statistic that deserves attention for its significance. While the loss function's application provides a visual representation of the strengths and weaknesses of the model's predictions, the value of the loss function changes when the data change, which means that the loss function is also stochastic. In such cases, one can only obtain a smaller set of models, called the model confidence set (MCS) [48], which contains the best forecasting model at a given level of confidence. The models in the MCS performs equally well at the given confidence level. Following Wang et al. [49] and Wang et al. [50], we construct the MCS test after out-of-sample prediction. For a given model, a larger "MCS p-value" means higher prediction accuracy. A very classical test for the significance of predictive accuracy, the Diebold-Mariano (DM) test [51], is also used in this paper. The null hypothesis of the DM test is that there is no difference in the two models' predictive accuracy, and the smaller the p-value, the smaller the probability of erroneously rejecting this null hypothesis. The DM test and MCS test are used to prove the superiority of the proposed model statistically. For details of these two test methods, see Appendix A.2 and A.3.

4. Data

The daily price of EUA futures is used for empirical testing [6,1] which is obtained from the *Wind* database. As shown earlier, an

Table 2
Carbon price sample size and date range.

		Sample size	Date range
Case I	Sample set	258	2017.11.01–2018.10.31
	Training set	233	2017.11.01–2018.09.26
	Test set	25	2018.09.27–2018.10.31
Case II	Sample set	258	2018.11.01–2019.10.31
	Training set	233	2018.11.01–2019.09.26
	Test set	25	2019.09.27–2019.10.31



(a) First decomposition result of Case I (2017.11.1-2018.9.26)

(b) First decomposition result of Case II (2018.11.1-2019.9.26)

Fig. 4. First decomposition result of carbon price.

important takeaway is that the core reforms of EU ETS Phase III are MSR as well as LRP. since the strengthening of MSR with increased LRF levels in 2017, EUA prices have skyrocketed, creating a significant difference from the depressed levels before 2017. Despite the MSR program's first appearance in 2015, the EUA price did not exceed €10. After 2017, following the decision to commit to strengthening the MSR and increasing the LRF, the EUA price steadily increased, peaking at over €25. Therefore, we focus mainly on Phase III and select the past two years before starting this study, namely, 11/2017–10/2019, as the sample interval. This provides insight into risk management for carbon market participants at this stage. This feature is fully reflected in Fig. 3, whereby current EU ETS market characteristics are unlike anything investors have experienced before 2017.

The sample is divided into two sections, Case I and Case II, to validate the model's carbon price prediction ability under different market scenarios. In Fig. 3, the characteristics of the two cases are discrepant. We observe a more significant upward trend in Case I (2017.11.01–2018.10.31), and Case II (2018.11.01–2019.10.31) has more substantial volatility characteristics.[6] Within each Case, the ratio of in-sample training data (training set) to out-of-sample predictive data (test set) is 10 to 1. The corresponding sample size and date range are presented in Table 2. Both of these cases are rarely encountered before 2017 and will be the norm for carbon market prices in the late 2020s and during future EU ETS Phase IV as the MSR and LRF programs continue to be implemented.

5. Empirical study

5.1. Decomposition of the carbon price

Because of the non-stationary and non-linear characteristics of the carbon price series, we use the adaptive and nonrecursive decomposition method VMD. Compared with wavelet decomposition and EMD, VMD exhibits its superiority to denoise and separate similar frequencies when managing the complex dynamics of data.

To avoid the data leakage problem that test set information is known to the training set during pattern decomposition, we do not follow studies that decompose the entire sample set (training set and test set); instead, we decompose the new training set before making new predictions, in other words, Case I and Case II must each be decomposed as many times as the out-of-sample predictions. Due to space limitations, Fig. 4 only shows the first decomposition result of the two cases. All the modes are arranged in order from high frequency to low frequency. The high-frequency modes have the characteristics of randomness, and low-frequency modes have strong periodic fluctuations, of which the last mode reflects the overall trend of the carbon price. Compared with the original carbon price series, decomposed modes show simple structures, stable fluctuations, and strong regularity, improving fitting and prediction accuracy. The analog signal frequency of each mode is presented in Table 3, and the corresponding time scale (period) is also calculated.

To compare VMD with EMD, we also implement the carbon price forecasting model based on EMD. The results show that VMD can effectively solve the inherent shortages of EMD, namely, end effect and mode mixing, to obtain more accurate predictions. Details of the comparison are explained in Section 4.3.

[6] See Appendix A.1 for the nonstationary and nonlinear tests of the two sample cases.

Table 3
Analog signal frequency and corresponding time scale of each mode.

		Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	Mode 6	Mode 7
Case I	Frequency (Hz)	0.361 ~ 0.391	0.33 ~ 0.352	0.232 ~ 0.24	0.155 ~ 0.193	0.116 ~ 0.146	0.094 ~ 0.099	0 ~ 0.043
	Period (Days)	2.56 ~ 2.774	2.841 ~ 3.026	4.161 ~ 4.315	5.178 ~ 6.472	6.853 ~ 8.63	10.13 ~ 10.591	23.3 ~ Inf
Case II	Frequency (Hz)	0.433 ~ 0.459	0.313 ~ 0.356	0.24 ~ 0.27	0.137 ~ 0.172	0.064 ~ 0.116	0.026 ~ 0.039	0 ~ 0.043
	Period (Days)	2.178 ~ 2.307	2.807 ~ 3.192	3.698 ~ 4.161	5.825 ~ 7.281	8.63 ~ 15.533	25.889 ~ 38.833	23.3 ~ Inf

Note: The mirror of the left half of the signal is taken, thus the frequency is less than 0.5.

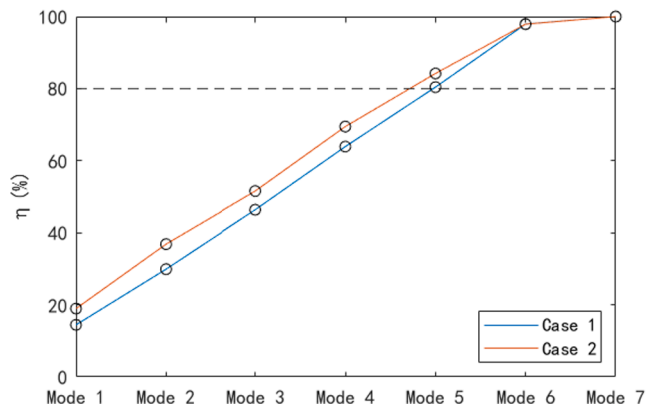


Fig. 5. High-frequency and low-frequency identification of the first decomposition.

Besides, in Fig. 5, mode 7 has a low complexity, which is consistent with the conclusion of Tang et al. [52] that the complexity of the trend term is low. Thus, the low-frequency component can be regarded as the carbon price's long-term trend after removing irregular noise. By contrast, the high-frequency modes have low variance contribution rates, with the sum less than 5%, and no significant trends, showing that they have a weaker interpretation of the original series and contain some random information.

The analysis above indicates that the carbon price exhibits various characteristics among high-frequency and low-frequency components, which may be due to multiple drivers' coupling. As a type of commodity market, the carbon market's price fluctuations are affected by internal market mechanisms (e.g., supply and demand mechanisms and price mechanisms). These random disturbances change rapidly, thus having little effect on the long-term trend of the carbon price. Meanwhile, the carbon market also has unique features. Nonmarket factors, namely, the external environmental factors of the carbon market, such as temperature, policies, energy price, and special events (e.g., economic crises and political changes), have primarily interfered with the carbon market mechanism [17] and caused long-term carbon price fluctuations.

Table 4
Variance contribution rate analysis and Mann–Kendall tendency test.

		Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	Mode 6	Mode 7
Case I	Rate	0.057%	0.052%	0.066%	0.089%	0.456%	0.268%	95.915%
	Tendency	0.9752	0.6169	0.9096	0.8671	0.9712	0.9832	<2.2e-16
Case II	Rate	0.498%	0.452%	0.616%	0.760%	2.204%	7.864%	73.467%
	Tendency	0.7808	0.6192	0.9658	0.9377	0.907	0.907	<2.2e-16

Note: The variance contribution rate formula is the variance of each mode/variance of the original series. The tendency rows list the p values of the Mann–Kendall tendency test. Rejecting the null hypothesis indicates that the tendency is significant.

5.2. Recognition of mode characteristics

The Lempel–Ziv complexity algorithm[7] is used to identify high-frequency and low-frequency components automatically. We apply the algorithm to modes from the first-time decomposition of two sample cases and present the identification results in Fig. 5. Case I does not reach the critical value until mode 5, where the value of η is 80.4%. Therefore, modes 1–5 are identified as high frequencies, and modes 6 and 7 are low frequencies. Similarly, in Case II, modes 1–5 are high frequencies, and modes 6 and 7 are low frequencies.

In Fig. 4, the random fluctuation of the high-frequency component is evident, and the low-frequency component, namely, modes 6 and 7, numerically account for a larger proportion of the original series. The variance contribution rate analysis and Mann–Kendall tendency test are conducted for all modes (Table 4). The results show that mode 7 explains the most fluctuations in the original series, with the variance contribution rate up to 95.915% in Case I, indicating that it contains the most important influencing factors of the carbon price change. Among other modes, the variance contribution rate of mode 6 is relatively large, reaching 7.864% in Case II. According to the Mann–Kendall tendency test, only mode 7 has a highly significant tendency in both cases.

5.3. One-step-ahead forecasting

According to the analysis mentioned above of the data characteristics inherent in different modes, the high-frequency component is determined by the market mechanism; thus, it is affected by market sentiment and has volatility clustering, which can be well characterized by the econometric model GARCH. For non-linear low-frequency modes, the application of econometric models is limited. Therefore, more flexible fitting functions should be considered, namely, artificial intelligence models with self-organization features, self-learning, and adaptability. Because the low-frequency component reflects long-term carbon price fluctuations, the LSTM neural network with long-term memory is used for modeling. This study proposes a GARCH and LSTM hybrid model, based on the decomposition-ensemble framework and the non-linear ensemble method, namely, a VMD-GARCH/LSTM-LSTM model, for carbon price forecasting. The one-step-ahead real-time prediction strategy is adopted here, namely, after predicting the price of a day, the real price of this day is appended to the training set for the next prediction's convenience.

5.3.1. Model selection and parameter description

To confirm the proposed model's superiority, it is compared with various carbon price forecasting models selected based on the following considerations. For one thing, to examine differences between

[7] Before using this algorithm, the input sequence needs to be converted into a binary sequence. In this paper, the median of the sequence is set as the threshold. If the number exceeds the threshold, it is transformed to 1; otherwise, it is transformed to 0. In this study, the critical value λ_0 is set to 80%.

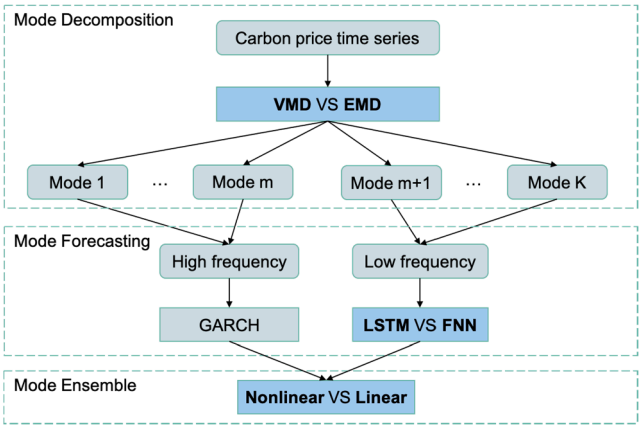


Fig. 6. Methods to be compared in each stage of the forecasting process.

prediction performance of the decomposition–ensemble model and the single model, a single LSTM forecasting model is included for comparison. For another, to examine whether the techniques employed in our proposed model are more effective than commonly used methods in a decomposition–ensemble model, we run through these methods in every possible combination. As shown in Fig. 6, in each stage of the multi-scale non-linear ensemble forecasting process, there are two methods to be compared with each other. Specifically, in the mode decomposition stage, either VMD or EMD can be employed to decompose the carbon price time series; in the mode forecasting stage, both LSTM and FNN are available to forecast low-frequency modes; in the mode ensemble stage, we can select between non-linear and linear ensemble methods to combine forecasts of all the modes. As a result, the following models are obtained: VMD-GARCH/LSTM-ADD, VMD-GARCH/LSTM-LSTM (our proposed model), VMD-GARCH/FNN-ADD, VMD-GARCH/FNN-FNN, EMD-GARCH/LSTM-ADD, EMD-GARCH/LSTM-LSTM, EMD-GARCH/FNN-ADD, EMD-GARCH/FNN-FNN.

Notably, because the number of VMD modes must be determined by us rather than by the algorithm, we use the mode numbers obtained by the decomposition through EMD.[8] Before establishing the GARCH volatility model, we first establish an ARIMA mean model for the series and test the residuals' ARCH effect. If the ARCH effect exists, we jointly estimate the ARIMA and GARCH(1,1) models[9]. Regarding order determination of the ARIMA model, the difference order (d) is determined by the stationarity test, and then the optimal numbers of autoregressive terms (p) and moving average terms (q) are selected within the range of no more than three[10] through the information criterion BIC.[11]

For the LSTM model parameter setting, a sequence input layer, an LSTM layer, a fully connected layer, and a regression layer are included. The numbers of features and responses are specified as 1, and the number of hidden units in the LSTM layer as 200. To restrain the exploding gradient, the gradient threshold is set to 1. A total of 250 epochs are used to train, and the initial learning rate is specified as 0.005, which is reduced by multiplying a factor of 0.2 after 125 epochs. Notably, when building a neural network model, it is critical to choose the best optimizer, which helps the model converge quickly and learn correctly. Adam is a first-order optimization algorithm that can replace the traditional stochastic gradient descent process. Adam works well in practical applications and surpasses other adaptive technologies. Therefore, this study uses Adam optimizer to iteratively update the neural network weights based on the training data. Besides, to ensure the models' comparability, all LSTM models use the same parameter setting, and FNN models maintain the same parameters as LSTM except that the LSTM layer is changed to a fully connected layer[12].

5.3.2. Prediction evaluation and test results

Fig. 7 shows the out-of-sample prediction results for Case I and Case II by all the models above. It is clear that compared with other models, predicted values of VMD-GARCH/LSTM-LSTM have a higher degree of accuracy to real values of the carbon price. The accuracy lies in both quantity and direction; in other words, there are both a smaller quantitative difference and a more consistent upward or downward trend between the predicted value and the real value at each time point.

To further compare the forecasting accuracy of different models, out-of-sample prediction errors for two cases under three criteria (RMSE, MAE, MAPE) are presented in Fig. 8. It can be found that the prediction error of a single LSTM model is greater than that of all the decomposition–ensemble models. This conclusion is valid for two cases and three evaluation criteria, proving the decomposition-ensemble method's effectiveness for carbon price prediction. The main reason is that after mode decomposition, the complex carbon price series is transformed into simple, stable, and regular mode structures, which significantly improves the prediction accuracy.

Table 5 shows the prediction performance ranks of different models. There is almost no change in the order of models according to different evaluation criteria within one Case. However, across two cases, the ranks of some models have changed. There are only three models whose ranks remain unchanged. One model is LSTM, which always lags behind all other models, reflecting the disadvantages of the single prediction model compared with the decomposition–ensemble model. The other two models are based on the VMD algorithm and the GARCH/LSTM prediction algorithm (VMD-GARCH/LSTM-ADD and VMD-GARCH/LSTM-LSTM), which remain in the top two. This finding shows the superiority of the combination of VMD and GARCH/LSTM, namely, whether adopting the linear or non-linear ensemble method, it is always ahead of other models.

In Case I, as the carbon price is rapidly increasing, information on long-term trends is very helpful for carbon price forecasting. Using a model alone to predict the carbon price during this phase may allow the short-term fluctuating component to infect the model's uptake of information about the long-term trend component. With a decomposition-ensemble approach, the different periods' information components in each sub-mode are better absorbed by the particular model. Such a finding extends the work of Atsalakis [2], Daskalakis et al. [39], Guðbrandsdóttir and Haraldsson [33], Fan et al. [18], and others using a single model for carbon price forecasting. The carbon price is more volatile in Case II as shown earlier, LSTM does not seem to work well for sequences with weaker trends. This finding supports the praise of Zhu et al. [28] and others for the application of the decomposition-ensemble method to carbon price prediction.

Among the remaining six models, four models only move one place up or down in prediction performance across two cases, and two models (EMD-GARCH/LSTM-ADD and EMD-GARCH/LSTM-LSTM) make huge changes. Specifically, EMD-GARCH/LSTM-ADD increases from 8th in Case I to 5th in Case II, and EMD-GARCH/LSTM-LSTM decreases from 3rd in Case I to 6th in Case II. Thus, we observe that combining the EMD algorithm and the GARCH/LSTM prediction algorithm is sensitive to different carbon price change patterns and its forecasting performance is

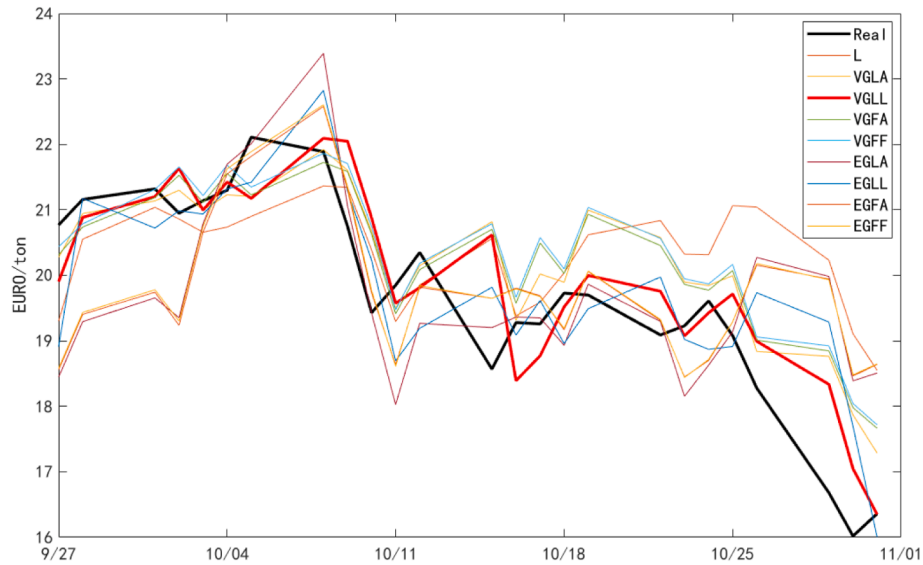
[8] In the many times of decomposition, the number of EMD modes is 7 first and then becomes 8. For simplicity, the number of VMD modes always takes 7.

[9] The order of the GARCH model is difficult to determine. In practical applications, only the low-order GARCH model is used [47]. In this study, GARCH (1,1) is selected.

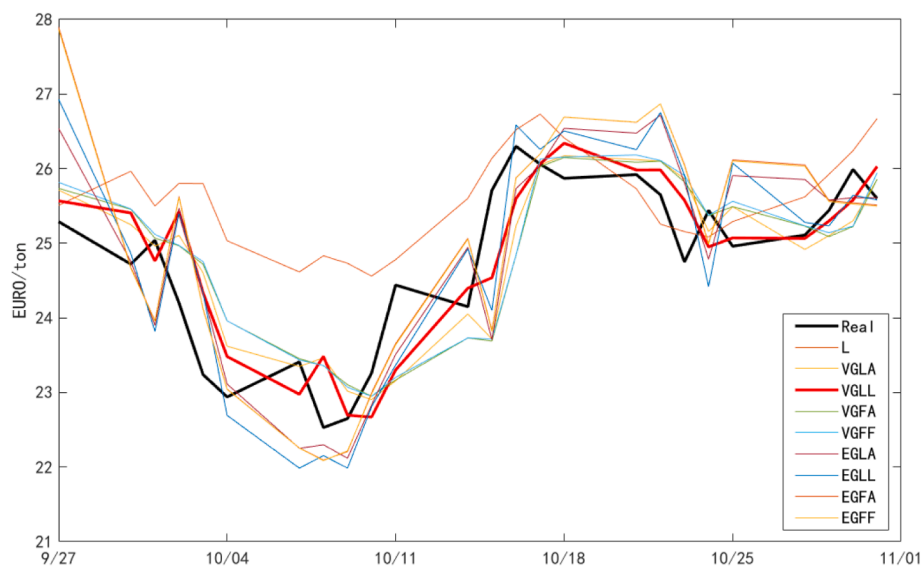
[10] To ensure the simplicity of the model, a high order is not suggested.

[11] Some scholars have used information criteria to choose the difference order (d), but the difference process changes the data used in the calculation of the likelihood function, which makes the BIC values of models with various difference orders unable to be compared; thus, this study uses information criteria to select only p and q values.

[12] The number of hidden units is still 200.



(a) Prediction results for Case I



(b) Prediction results for Case II

Fig. 7. Out-of-sample prediction results of different models. Note: L, VGLA, VGLL, VGFA, VGFF, EGLA, EGLL, EGFA, and EGFF are LSTM, VMD-GARCH/LSTM-ADD, VMD-GARCH/LSTM-LSTM, VMD-GARCH/FNN-ADD, VMD-GARCH/FNN-FNN, EMD-GARCH/LSTM-ADD, EMD-GARCH/LSTM-LSTM, EMD-GARCH/FNN-ADD, and EMD-GARCH/FNN-FNN, respectively. The same abbreviations are used below.

unstable. Especially for EMD-GARCH/LSTM-ADD, the RMSE difference between the two cases is close to 0.6, showing that the linear ensemble method does not have error correction ability compared to the non-linear ensemble method.

The VMD-GARCH/LSTM-LSTM model proposed in this study has the minimum error under all three evaluation criteria of RMSE, MAE, and MAPE, indicating that it well combines the advantages of the decomposition–ensemble technique, econometric and neural network hybrid modeling, and non-linear ensemble method. The model also achieves the best prediction performance in Cases 1 and 2, showing its adaptability to different carbon price change patterns, which proves its robustness.

According to the MCS test results in Table 6, for two cases and six loss functions[13], only VMD-GARCH/LSTM-ADD and VMD-GARCH/LSTM-LSTM keep in $\widehat{M}^{*}_{75\%}$ and the MCS p-value of VMD-GARCH/LSTM-LSTM is always 1. This finding further verifies that the multi-scale hybrid model combining VMD and GARCH/LSTM is more robust than the other models. According to the DM test results in Table 7, when the

[13] It should be noted that the loss funcitons used here are not the same with the prediction evaluation criteria mentioned above. This is due to concerns about robustness, i.e., more indicators adopted can increase the credibility of results.

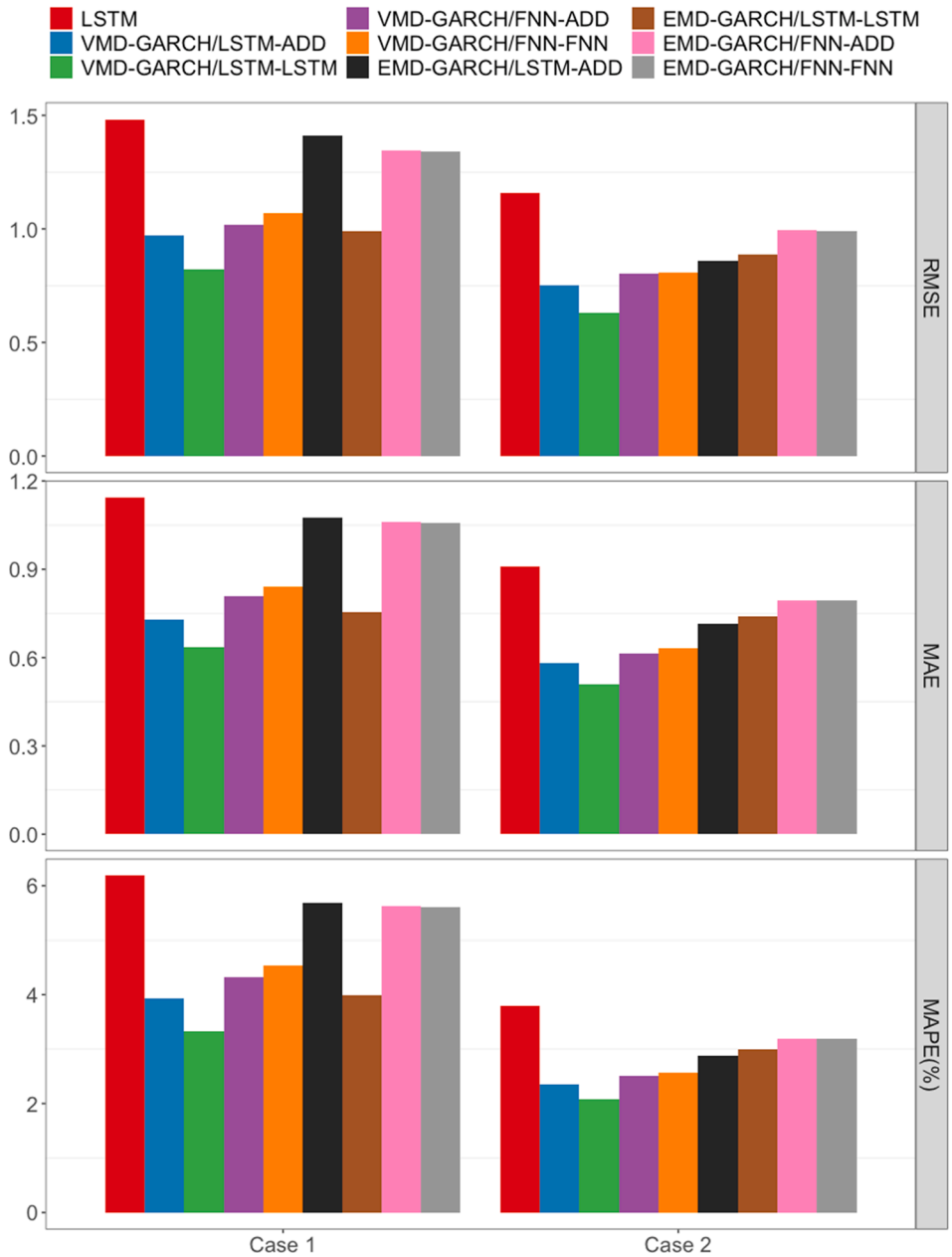


Fig. 8. Performance of different models under three criteria.

Table 5
Prediction performance ranks of different models.

Model	Case I			Case II		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE
LSTM	9	9	9	9	9	9
VMD-GARCH/LSTM-ADD	2	2	2	2	2	2
VMD-GARCH/LSTM-LSTM	1	1	1	1	1	1
VMD-GARCH/FNN-ADD	4	4	4	3	3	3
VMD-GARCH/FNN-FNN	5	5	5	4	4	4
EMD-GARCH/LSTM-ADD	8	8	8	5	5	5
EMD-GARCH/LSTM-LSTM	3	3	3	6	6	6
EMD-GARCH/FNN-ADD	7	7	7	8	7	8
EMD-GARCH/FNN-FNN	6	6	6	7	7	7

Note: For MAE, EMD-GARCH/FNN-ADD and EMD-GARCH/FNN-FNN are equal 7th under this study's data precision.

single LSTM model is the target model and other models are benchmark models, most p values are less than the significance level of 0.1, and the statistics are greater than 0, indicating that the multi-scale prediction model is significantly better than the single prediction model. Additionally, the DM test also confirms that the VMD-GARCH/LSTM-LSTM model has better carbon price prediction performance than other models; almost all the p values are less than the significance level of 0.1 when VMD-GARCH/LSTM-LSTM is the target model.

Based on the above statistical test results, it can be safely concluded that the VMD-GARCH/LSTM-LSTM model proposed in this study has significant advantages in capturing and predicting the fluctuation characteristics of the carbon price.

Table 6
MCS test results.

	Model	Loss function					
		MSE1	MSE2	QLIKE	R2LOG	MAE1	MAE2
Case I	L	0.1240	0.1222	0.1504	0.1350	0.0354	0.0318
	VGLA	1.0000*	1.0000*	1.0000*	0.4902*	1.0000*	1.0000*
	VGLL	1.0000*	1.0000*	1.0000*	1.0000*	1.0000*	1.0000*
	VGFA	1.0000*	1.0000*	0.6546*	0.3408*	0.0132	0.0182
	VGFF	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	EGLA	0.0708	0.0636	0.0724	0.0980	0.9950*	0.9976*
	EGLL	1.0000*	1.0000*	1.0000*	0.8064*	1.0000*	1.0000*
	EGFA	0.7114*	0.7832*	0.9010*	0.3644*	0.9950*	1.0000*
	EGFF	0.7180*	0.7832*	1.0000*	0.3704*	1.0000*	0.9976*
Case II	L	0.3628*	0.3216*	0.4080*	0.4046*	0.0316	0.0288
	VGLA	1.0000*	1.0000*	1.0000*	0.5540*	1.0000*	1.0000*
	VGLL	1.0000*	1.0000*	1.0000*	1.0000*	1.0000*	1.0000*
	VGFA	0.3332*	1.0000*	0.3418*	0.3358*	1.0000*	1.0000*
	VGFF	0.0704	0.0614	0.1056	0.1020	0.3946*	0.2918*
	EGLA	0.0010	0.0010	0.0012	0.0004	0.0032	0.0040
	EGLL	0.0000	0.0002	0.0000	0.0000	0.0002	0.0006
	EGFA	0.0692	0.1026	0.0324	0.0588	0.0074	0.0096
	EGFF	0.0626	0.0930	0.0298	0.0518	0.0068	0.0090

Note: This table lists the MCS p-value of the forecasting model. * means that the p-value is more than 0.25, namely, the model is in $\widehat{M}^{*}_{75\%}$.

Table 7
DM test results.

	Target model	Benchmark model							
		VGLA	VGLL	VGFA	VGFF	EGLA	EGLL	EGFA	EGFF
Case I	L	2.486(0.02*)	2.551 (0.018*)	2.419(0.024*)	2.215(0.037*)	0.476(0.638)	2.649(0.014*)	0.965(0.344)	1.023(0.316)
	VGLA		1.598(0.123)	−1.462(0.157)	−2.896 (0.008*)	−2.076 (0.049*)	−0.123(0.903)	−1.871 (0.074*)	−1.856 (0.076*)
	VGLL			−1.941 (0.064*)	−2.326 (0.029*)	−2.495(0.02*)	−1.104(0.28)	−2.28(0.032*)	−2.264 (0.033*)
	VGFA				−3.63(0.001*)	−2.001 (0.057*)	0.215(0.831)	−1.794 (0.086*)	−1.779 (0.088*)
	VGFF					−1.729 (0.097*)	0.589(0.561)	−1.513(0.143)	−1.496(0.148)
	EGLA						3.171(0.004*)	1.402(0.174)	1.532(0.139)
	EGLL							−2.588 (0.016*)	−2.572 (0.017*)
	EGFA								1.008(0.324)
Case II	L	1.967 (0.061*)	2.668 (0.013*)	1.766(0.09*)	1.742(0.094*)	1.394(0.176)	1.286(0.211)	0.674(0.507)	0.692(0.496)
	VGLA		1.437(0.164)	−1.552(0.134)	−1.797 (0.085*)	−1.319(0.2)	−1.231(0.23)	−1.426(0.167)	−1.432(0.165)
	VGLL			−1.757 (0.092*)	−1.874 (0.073*)	−2.292 (0.031*)	−2.454 (0.022*)	−2.012 (0.056*)	−2.024 (0.054*)
	VGFA				−1.112(0.277)	−0.599(0.555)	−0.706(0.487)	−1.099(0.283)	−1.097(0.283)
	VGFF					−0.519(0.609)	−0.651(0.521)	−1.068(0.296)	−1.065(0.297)
	EGLA						−0.566(0.576)	−1.173(0.252)	−1.176(0.251)
	EGLL							−1.093(0.285)	−1.088(0.287)
	EGFA								0.96(0.347)

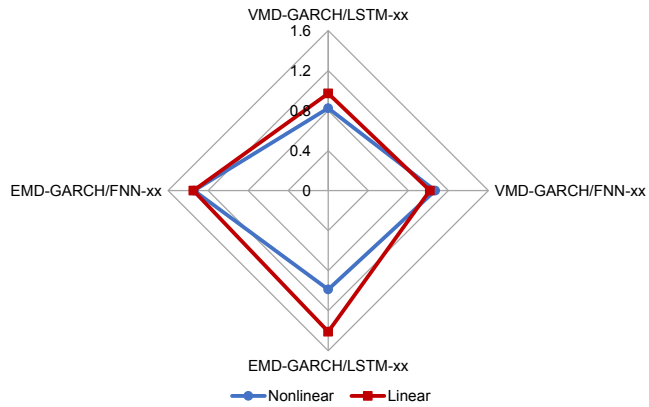
Note: * means that the p-value is less than 0.1.

5.3.3. *Performance comparison and analysis*

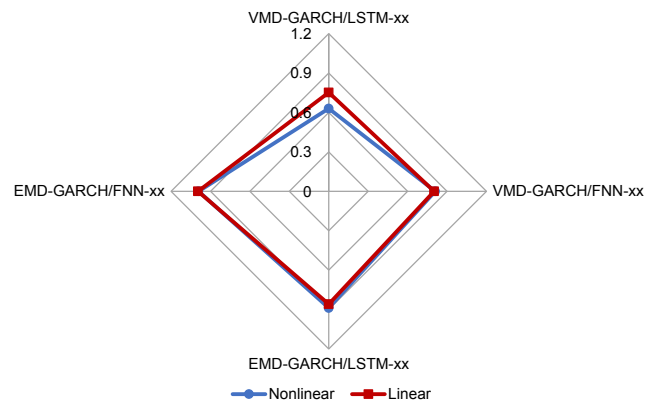
Here we would further compare our VMD-GARCH/LSTM-LSTM with other benchmark models. In Fig. 9[14], we compare the non-linear ensemble and linear ensemble models, finding that the non-linear ensemble method improves prediction accuracy. In contrast to Lu et al. [30], Hao and Tian [29], Tian and Hao [4], and other decomposition–ensemble forecast studies, which only use a simple summation model when performing prediction value integration, our findings reveal the advantages of using LSTM to integrate subsequence prediction values. VMD-GARCH/LSTM-LSTM is remarkably better than VMD-GARCH/LSTM-ADD in both cases. In Case I, the RMSE of EMD-GARCH/LSTM-LSTM is smaller than that of EMD-GARCH/LSTM-ADD in nearly 0.5. The reason for this finding is that the decomposition–ensemble model predicts each mode separately, leading to the error accumulation when integrating the prediction result of each mode, which can be overcome by a non-linear ensemble. Therefore, the non-linear ensemble algorithm can improve the hybrid model's prediction performance, and when GARCH and LSTM are used as prediction algorithms, the promotion is especially significant.

In Fig. 10, models based on VMD generally have smaller errors than

[14] Because the models have similar prediction performance ranks under the three criteria of RMSE, MAE, and MAPE, only the RMSE results are used in the following analysis.

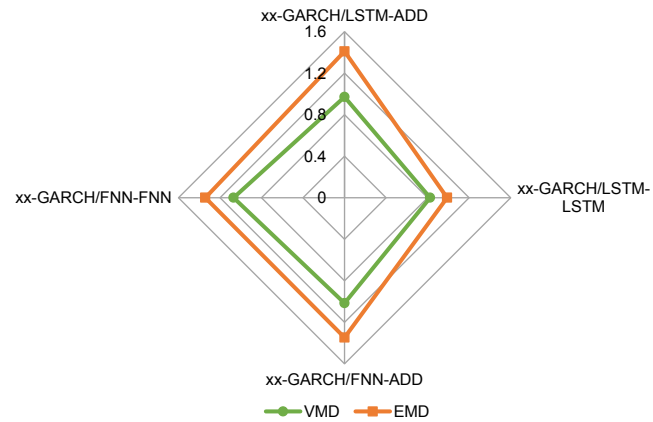


(a) RMSE performance of models in Case I

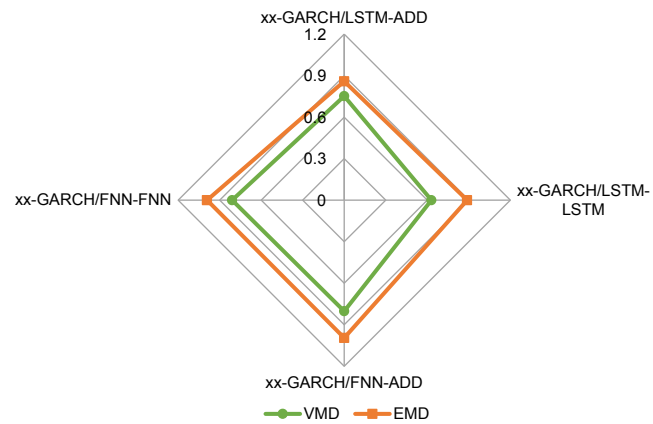


(b) RMSE performance of models in Case II

Fig. 9. Comparison of non-linear and linear ensemble models. Note: "xx" in the abbreviation means that selecting methods in this stage is being focused on, Figs. 10 and 11 the same. Especially for Fig. 9, "linear" corresponds to "ADD" and "non-linear" corresponds to the ensemble method, which is the same as the neural network for prediction.



(a) RMSE performance of models in Case I



(b) RMSE performance of models in Case II

Fig. 10. Comparison of VMD and EMD models.

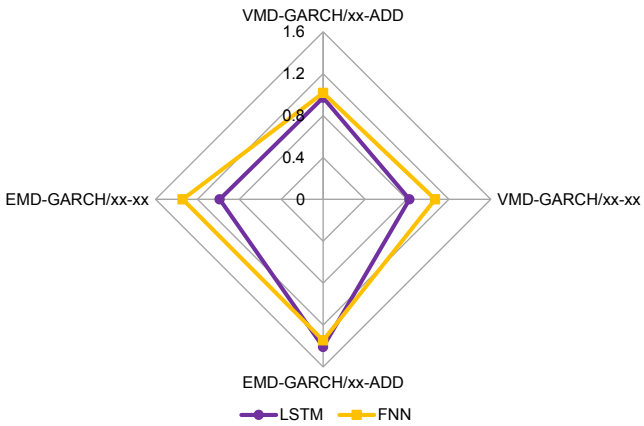
models based on EMD. For example, in Case II, even the VMD model with the largest error (VMD-GARCH/FNN-FNN) is superior to the EMD model with the smallest error (EMD-GARCH/LSTM-ADD). The main reason for this finding is that VMD overcomes the EMD defects of end effect and mode mixing, and can decompose the carbon price series more accurately. This result shows that the VMD algorithm has a significant advantage in improving the performance of the carbon price prediction model. This finding corroborates the reviews of Sun et al. [12] and Tian and Hao [4] on the excellent performance of VMD in carbon price forecasting. Compared to Sun et al. [12] and Tian and Hao, we more comprehensively apply both metrics and machine learning models in the prediction session and adopt a non-linear (LSTM) approach in the predictive value integration part.

At the level of application of the decomposition method, our VMD-GARCH/LSTM-LSTM extends Li and Lu [25], Zhu [13], Zhu et al. [6] and Zhu et al. [26], which apply the EMD method for decomposition. The excellent performance of VMD is fully reflected in this study. The use of VMD rather than EMD for carbon price prediction effectively improves the prediction accuracy in the more complex temporal and spatial context of carbon price fluctuations in this study.

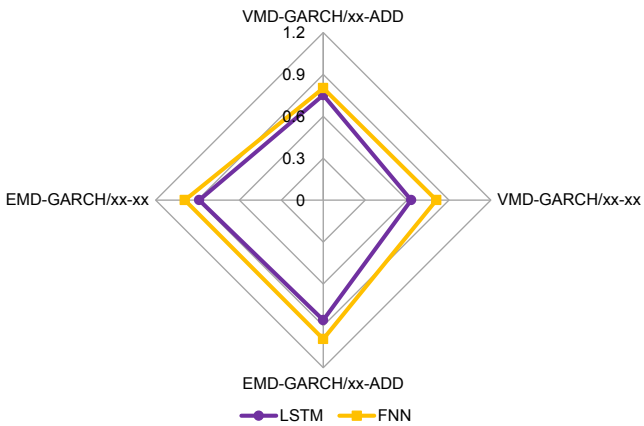
In Fig. 11, models based on LSTM are generally more accurate than decomposition-ensemble models based on FNN [18,4,13], which is more obvious when the non-linear ensemble method is adopted. For example, in Case I, the prediction error of VMD-GARCH/LSTM-LSTM is more than 0.2 lower than that of VMD-GARCH/FNN-FNN, and the prediction error of EMD-GARCH/LSTM-LSTM is nearly 0.4 lower than that of EMD-GARCH/FNN-FNN. This finding is that FNN is inadequate for learning time series; it requires much accompanying treatment while the performance is not necessarily improved. For carbon price prediction tasks where the information incorporates complex time correlations, especially low-frequency modes that reflect long-term carbon price fluctuations, using LSTM with long-term memory is a better choice.

Compared to Fan et al. [18], Tian and Hao [4], and Zhu [13], this finding seems to imply that the LSTM is better suited to adapt to the changing form of the carbon price in a more complex period of EU ETS carbon price variation. As the carbon price shows rapid growth after 2017, the long-term trend is essential for current carbon price forecasts, and the LSTM takes full advantage of the long-term trend in a way that the ANN cannot.

This evidence also implies that models such as the SVM may not be able to make fair use of information on the long-term rise in carbon market prices for carbon price forecasting. Although the work of Zhu and Wei [1], Zhu et al. [26], and Lu et al. [30] has significant contributions, it has not yet been possible to determine whether this predictive accuracy will remain profitable in the face of the new market scenario of the EU ETS.



(a) RMSE performance of models in Case I



(b) RMSE performance of models in Case II

Fig. 11. Comparison of LSTM and FNN models.

5.4. Multisteps-ahead forecasting

The analysis as mentioned earlier only focuses on one-step-ahead forecasting. However, sometimes it matters that multistep-ahead forecasts for carbon prices are made in high accuracy. Multistep-ahead forecasting, specifically, puts the one-step-ahead forecast into historical data as a "real" value and then uses this new data set to predict the next-step value. The multistep-ahead forecasting process leads to error accumulation, which is more prominent within the decomposition–ensemble framework. Thus, as a robustness test, we compare the multistep-ahead forecasting performance of the decomposition–ensemble models in this section.

Fig. 12 shows the multistep-ahead forecasting errors when changing the time step among 1, 2, 4, and 6. As the time step increases, all the models' errors display an ascending trend, which is a reflection of the error accumulation problem. Besides, there is an interesting finding that multistep-ahead forecasting performance of decomposition–ensemble models are highly correlated with the decomposition method, especially when the time step is in a relatively large value. As shown in the Fig. 12, when making four-step-ahead and six-step-ahead forecasts, the EMD-based models' error curves become incredibly steep. By contrast, the VMD-based models have much gentler error curves, with the maximum RMSE value less than 2.5.

The VMD-GARCH/LSTM-LSTM model has significantly defeated the EMD-based decomposition-ensemble model compared to existing models with increasing prediction accuracy ([1,26;30]). VMD can better extract the long-term carbon price trend, while EMD easily allows the long-term carbon price trend to retain the short-term trend's redundant component. As the EU ETS market shows a long-term upward trend after 2017, improving the utilization of long-term price information becomes crucial for forecasting accuracy. As the number of forecast steps grows, the disadvantage with non-LSTM forecasts [2,18,19] becomes apparent, which may likewise be due to the inefficient utilization of long-term price trend information by other forecasting methods.

In sum, VMD can significantly improve the accuracy of multistep-ahead forecasts in the proposed decomposition–ensemble paradigm, while the selection of neural networks and ensemble methods have little to do with the forecasting performance. Compared with other models, VMD-GARCH/LSTM-LSTM has a lower and more stable error curve, further showing its robustness in carbon price forecasting.

5.5. Sensitivity of the parameter

Neural networks (LSTM or FNN) are used in both our proposed model and the benchmark model in this study, and their performance is easily influenced by their hyperparameters, especially the number of hidden units. Thus, this section investigates to what extent this parameter changes the prediction accuracy of various models.

Fig. 13 shows each model's RMSE distribution when its parameter of neural network hidden unit number is changed.[15] As mentioned earlier, the original number of hidden units is set to 200; based on this, we experiment on the parameter values of 50, 100, 150, 200, and 300, forming a box containing six RMSE values.

The single LSTM model box is the longest, which is much longer than the other boxes, implying that the single model's disadvantages compared with the decomposition–ensemble model are reflected in prediction accuracy and insensitivity the parameter setting. This finding supports the importance of our integrated use of econometric and AI methods and adopting a decomposition-ensemble approach. In contrast to Xu et al. [19], Atsalakis [2], and Koop and Tole [11], the combined use of different models can simultaneously exploit the unique advantages of each model for a particular data feature.

Besides, it found that all the FNN-based models are stable, the boxes are close to lines; compared with them, LSTM-based models are more sensitive, with the box having a certain length. However, in terms of average or median results, LSTM-based models are relatively lower than FNN-based models, showing that the former has higher prediction accuracy than the latter. This finding indicates that when choosing between LSTM-based and FNN-based models, we should balance accuracy against sensitivity. Considering that although LSTM-based models are less stable, the range of prediction errors is within an acceptable range, e.g., among the four models employing VMD as the decomposition algorithm, even the upper bounds of VMD-GARCH/LSTM-ADD and VMD-GARCH/LSTM-LSTM are lower than those of VMD-GARCH/FNN-ADD and VMD-GARCH/FNN-FNN, our proposed model is based on LSTM.

The VMD-GARCH/LSTM-LSTM model proposed in this study has a significantly lower average or median than other models and a within-an-acceptable-range sensitivity to the parameter setting; thus, it is fair to say that this model exhibits a relatively better performance in carbon price forecasting.

[15] Due to space limitations, we only analyze the situation of one-step-ahead forecasting for Case I and use the criterion of RMSE here.

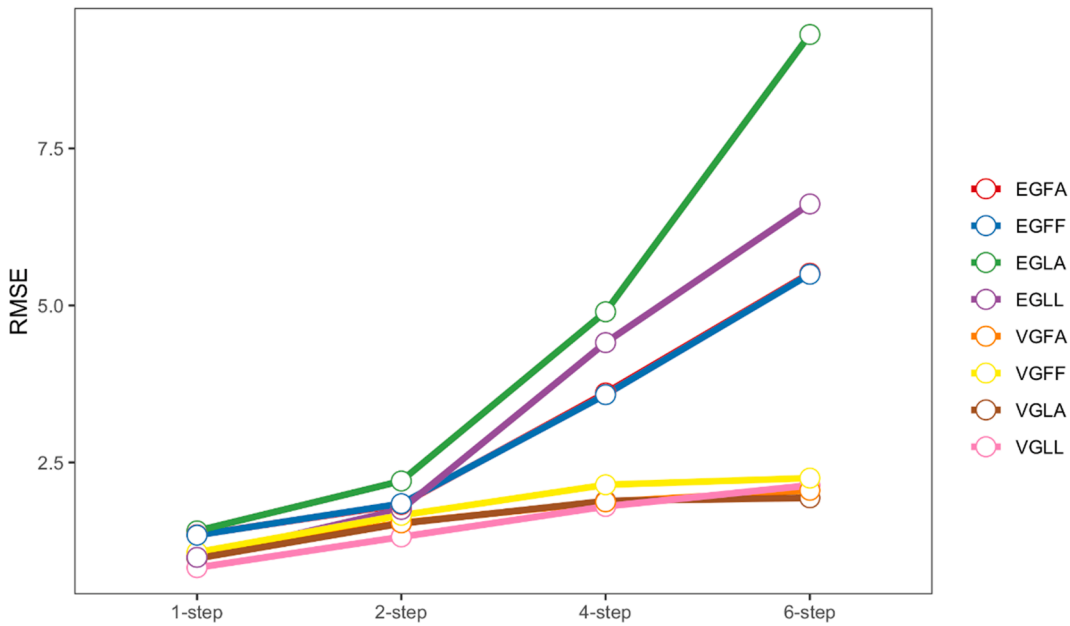


Fig. 12. Multistep-ahead forecasting errors.

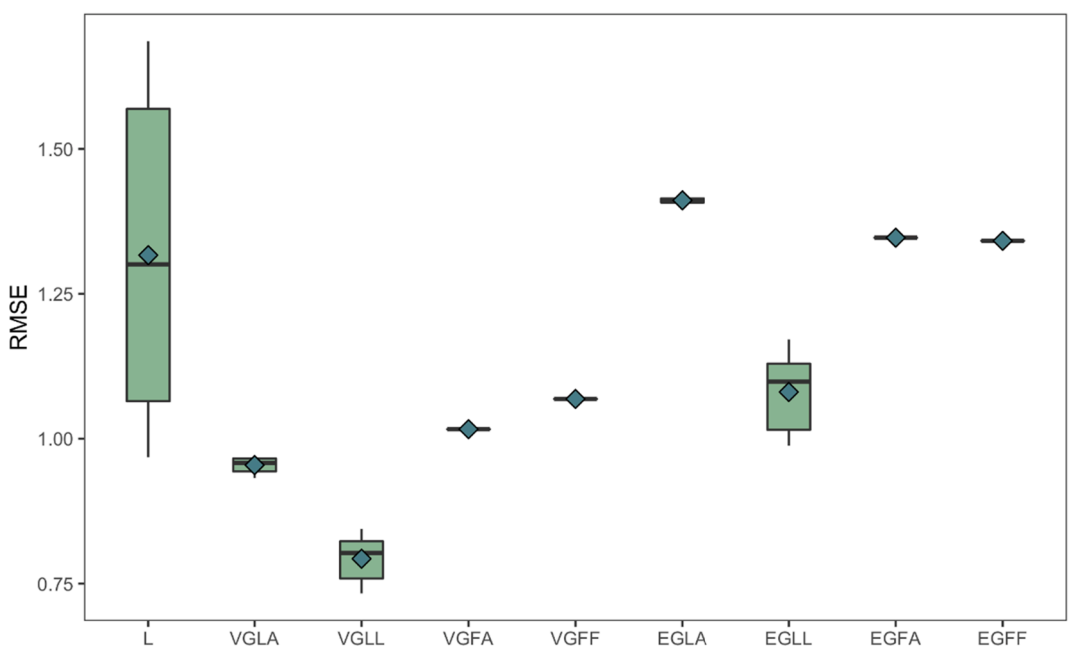


Fig. 13. RMSE distribution of each model. Note: Other than the standard notations, we use the diamond-shaped marker to denote the average. According to the boxplot definition, a longer box indicates that this model's performance is more sensitive to the parameter setting.

6. Conclusion

After 2017, there is a stunning new change in the EU ETS Phase III carbon price. This new change exposes carbon market investors to unprecedented price risk and has triggered a widespread demand for carbon price forecasts. This study proposes a novel hybrid model VMD-GARCH/LSTM-LSTM for carbon price prediction based on the decomposition-ensemble strategy [21] and utilizes econometric or neural network forecasting methods according to mode characteristics, with forecasts of all modes combined by the non-linear ensemble method. As the EU ETS carbon price has shown a significant upward trend and intense volatility over the current period, our model can better adapt to the current price characteristics to improve forecast accuracy. Using the late EU ETS Phase III data for empirical analysis, it proves that the proposed VMD-GARCH/LSTM-LSTM model has remarkable advantages in forecasting highly non-stationary and non-linear carbon price series during late EU ETS Phase III. The following main findings are obtained.

(1) The prediction error of the single model is greater than all decomposition-ensemble models. The philosophy of the decomposition-ensemble method is effective in the field of carbon price prediction. Also, the non-linear ensemble algorithm can improve the prediction performance of the multi-scale model. When GARCH and LSTM are used as the prediction algorithms, the improvement is especially noticeable.

(2) Prediction errors of models based on the VMD algorithm are significantly smaller than those based on EMD. The VMD algorithm has a significant advantage in improving the model's carbon price prediction performance, especially in multistep-ahead forecasting. Models based on LSTM are generally more accurate than models based on FNN, which is more evident when the non-linear ensemble method is used.
(3) Whether the combination of VMD and GARCH/LSTM adopts the linear or non-linear ensemble method, it is always ahead of the other models. However, the combination of EMD and GARCH/LSTM is sensitive to different carbon price change patterns and thus does not have stable carbon price prediction performance.

As the MSR will be more intensively applied to the EU ETS during phase IV [53], the EUA price change may enter a new upward or downward path to better serve as a ETS market guide. Recent contributions by Richstein et al. [54], Perino and Willner [9] and Beck and Kruse-Andersen [55] evaluate the impact of the MSR on price paths. Perino and Willner [9] and Richstein et al. [54] find that the MSR increases price volatility. Based on this fact and our research findings, the VMD-GARCH/LSTM-LSTM can be better aligned to scenarios of dramatic price changes in the EU ETS market today or overtime in the future. Further, with the increasing concern about carbon emission [56,57,58] and the rapid development of China's carbon market [59,60,61], our research also provides insights into carbon market price forecasts around the world to meet the needs of investors.

CRediT authorship contribution statement

Yumeng Huang: Methodology, Software, Writing - original draft. **Xingyu Dai:** Methodology, Data curation, Writing - review & editing. **Qunwei Wang:** Supervision, Writing - review & editing. **Dequn Zhou:** Supervision, Resources, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

Authors are grateful to the financial support from the National Natural Science Foundation of China (nos. 71922013, and 71834003).

Appendix A

A.1. Non-stationary and non-linear tests of the carbon price

The augmented Dicky–Fuller (ADF) test and Brock–Decher–Scheikman (BDS) test are widely used in examining non-stationary and non-linear characteristics of data, respectively. In this study, the ADF and BDS tests are performed on training sets of the two carbon price sample cases. The BDS test projects the series and the delay-order series into a vector space as a point. The first-order delay is two-dimensional, and the second-order delay is three-dimensional. Here, we first fit an ARIMA model to the series and then perform the BDS test on the residuals. The dimension is set to 2–5, and the closed-point epsilon value is set to 0.7 times the standard deviation of the tested data. The results of ADF and BDS tests are presented in Tables A1 and A2, respectively, and we conclude that the carbon price series is non-stationary and non-linear.

A.2. DM test

The DM test's null hypothesis is that the target model A and the benchmark model B have the same prediction accuracy, and the alternative hypothesis is that the two have different prediction capabilities. Using the mean square error as the loss function, the DM statistic is as follows:

$$S = \frac{\overline{g}}{\sqrt{\frac{\widehat{V_{\overline{g}}}}{M}}} \tag{A1}$$

where

Table A1
ADF test results.

	Statistic	P-value	Stationarity
Case I	−2.686	0.288	×
Case II	−2.718	0.274	×

Note: Lag order is 6.

Table A2
BDS test results.

	M-dimensional space								Linearity
	2		3		4		5		
	Statistic	P-value	Statistic	P-value	Statistic	P-value	Statistic	P-value	
Case I	5.6308	0.000	6.0538	0.000	6.4033	0.000	6.7378	0.000	×
Case II	−2.4245	0.015	−1.7210	0.085	−1.7835	0.074	−1.7314	0.083	×

Note: × indicates non-linearity at the significance level of 0.1.

$$\overline{g} = \frac{1}{n}\sum_{t=1}^{n}\left[\left(x(t) - \widehat{x}_A(t)\right)^2 - \left(x(t) - \widehat{x}_B(t)\right)^2\right] \tag{A2}$$

$$\widehat{V}_{\overline{g}} = \gamma_0 + 2\sum_{l=1}^{\infty}\gamma_l, \gamma_l = Cov(g_t, g_{t-l}) \tag{A3}$$

$\widehat{x}_A(t)$ and $\widehat{x}_B(t)$ represent the predicted values of the test model A and benchmark model B at time t, respectively.

The p-value and S statistic of the DM test reflect the difference in prediction accuracy between models A and B. Specifically, a p-value less than the significance level indicates that the prediction accuracy of the two models is significantly different, and a negative S statistic indicates that the prediction error of the target model is less than the benchmark model.

A.3. MCS test

Six different loss functions are used to examine the prediction accuracy of models, namely, $L_i(i = 1, 2, \cdots, 6)$. L_1 and L_2 are respectively the mean square error and its quadratic form. L_3 is estimated by the Gaussian likelihood function, and L_4 is similar to the Mincer–Zarnowitz regression value. L_5 and L_6 are the average absolute error and its quadratic form, respectively. The definitions of the loss functions are as follows:

$$L_1 : MSE_1 = \frac{1}{n}\sum_{t=1}^{n}[x(t) - \widehat{x}(t)]^2 \tag{A4}$$

$$L_2 : MSE_2 = \frac{1}{n}\sum_{t=1}^{n}\left[x(t)^2 - \widehat{x}(t)^2\right]^2 \tag{A5}$$

$$L_3 : QLIKE = \frac{1}{n}\sum_{t=1}^{n}\left[\ln(\widehat{x}(t))^2 + \left(\frac{x(t)}{\widehat{x}(t)}\right)^2\right] \tag{A6}$$

$$L_4 : R^2LOG = \frac{1}{n}\sum_{t=1}^{n}\left[\ln\left(\frac{x(t)}{\widehat{x}(t)}\right)\right]^2 \tag{A7}$$

$$L_5 : MAE_1 = \frac{1}{n}\sum_{t=1}^{n}|x(t) - \widehat{x}(t)| \tag{A8}$$

$$L_6 : MAE_2 = \frac{1}{n}\sum_{t=1}^{n}\left|x(t)^2 - \widehat{x}(t)^2\right| \tag{A9}$$

The MCS test assumes that all prediction models are in a set M_T, and the relative loss function of any two models (u and v) is defined as $Y_t^{i,uv} = L_t^{i,u} - L_t^{i,v}$. By performing a series of significance tests in M_T, the models with poor predictive ability are gradually removed. The null hypothesis H_0 is that both models have the same predictive power, which takes the form $H_0 : \mathrm{E}\left(Y_t^{i,uv}\right) = 0$. The scope statistic is defined as

$$T_R = \max_{u,v\in M_T}\frac{\left|\overline{Y}_t^{i,uv}\right|}{\sqrt{var\left(Y_t^{i,uv}\right)}} \tag{A10}$$

$$\overline{Y}_t^{i,uv} = \frac{1}{n}\sum_{t=1}^{n}Y_t^{i,uv} \tag{A11}$$

In the MCS test, the empirical distribution of T_R can be obtained by the Bootstrap method. In terms of the Bootstrap method's parameter setting, the number of re-sampling times is 5000, and the data block length is 3. For a given model $k(k \in M_T)$, a larger p-value means higher prediction accuracy.

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